

Application / AI agent

one import → one engine
(superset bundle)

Embedding model

transformers.js ·
ORT-Web · /v1

Completion LLM

OpenAI-compatible
extract · reflect

OPFS

navigator.storage
(snapshot store)

imports zengram-lite

wasm-bindgen binding layer (→target web) — JS glue + injected host imports · the only boundary crossing

JS glue + host imports

WebAssembly sandbox — one .wasm (zengram-wasm links zeta-wasm + framework) · wasm32-unknown-unknown · single thread

ZengramMemory — agentic-memory API

finished vectors

Zengram framework (this paper)

Knowledge tier

knowledge · embedding
recall (vector+FTS)
confidence · decay

Session tracking

session · turn
part · tool_call
provenance

Context assembly

6-phase
token budget
fingerprint

Bring-your-own-result bridges — rememberWithVector · extractWithFacts · reflectWithInsights

framework runs as tables + queries on the engine

Zeta engine core — shared with the server build; detailed in the zeta-lite report

Memory backend: in-linear-memory tables + catalog — every framework table lives here

Parser → Planner
→ Executor

Log-centric MVCC
SI · versioned rows

HNSW + FTS
indexes

Branching
fork timestamps

ZL

Snapshot codec

export / openFromSnapshot

snapshot blob (bytes)

extracted facts / insights

re-exported engine